

**AI-DRIVEN CONVERSATIONAL ASSISTANT FOR
LEASING COMPANY CUSTOMER CARE**

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Dissertation submitted in partial fulfillment of the requirements for the Bachelor of
Science (Hons) in Information Technology Specialized in Information Technology

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DECLARATION

I declare that this is my own work, and this dissertation does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Signature of the Supervisor:  Date: 25/04/2026

(Mrs. Suriyaa Kumari)

DEDICATION

This dissertation is dedicated to my supervisors, friends, group members and all those who provided unwavering support throughout this research journey.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisors for the invaluable guidance, constructive feedback, and continuous support throughout this research. I am deeply grateful to Sri Lanka Institute of Information Technology for providing the academic environment and resources necessary to complete this work.

I would also like to acknowledge the leasing company supervisor who generously provided access to their operational data, customer care records, and staff expertise, which were essential to the development and validation of the proposed system.

Special thanks are due to my fellow members who provided feedback during the development and testing phases. Finally, I am grateful to my group members and our supervisors for their patience, encouragement, and support throughout the entire duration of this research.

ABSTRACT

AI-Driven Conversational Assistant for Leasing Company Customer Care

Efficiency in customer service has emerged as one of the decisive factors in the functioning of leasing companies, given that clients tend to need information related to leasing agreements, company information, location of branches, and related services. Thus, the main aim of the research presented here is to develop an artificial intelligence conversational assistant intended to perform functions within the scope of leasing company customer care operations and incorporating the capacities of large language models (LLMs) and knowledge base retrieval system.

Three main interaction forms are supported by the solution developed as text-based, voice-based, and image-based communication. For handling text-based questions, the DeepSeek V3 service is used, whereas the Google Gemini 2.0 Flash model is utilized for handling the image-based queries. In the case of voice communications, the Deepgram model is implemented for converting speech to text and vice versa. Knowledge base data is stored in the Supabase PostgreSQL database with vector embeddings for conducting the searches based on the semantic similarity.

In the designed solution, the dual access system is introduced as both Admin and User portals are incorporated. Within the Admin portal, knowledge base data can be managed by authorized users, whereas the User interface involves the interactive conversation with the assistant. As a part of this design, the human customer care representative escalation is introduced. Inference on large language models is performed with the help of OpenRouter API.

Thus, this research has shown that a combination of RAG with multimodal LLMs can result in enhanced performance of customer care services in the leasing sector.

Keywords: Large Language Models (LLM), Retrieval-Augmented Generation (RAG), Conversational AI, Knowledge Base, Vector Embeddings, Voice Assistant, Leasing Customer Care

TABLE OF CONTENTS

Table of Contents

DECLARATION	1
DEDICATION	2
ACKNOWLEDGEMENT	3
ABSTRACT	4
TABLE OF CONTENTS	5
LIST OF FIGURES	7
LIST OF TABLES	8
LIST OF ABBREVIATIONS	9
LIST OF APPENDICES	10
1. INTRODUCTION	1
1.1 Background & Literature Survey	3
1.2 Research Gap	6
1.3 Research Problem	8
1.4 Research Objectives	9
1.4.1 Main Objectives	9
1.4.2 Specific Objectives	9
2. METHODOLOGY	11
2.1 System Overview	14
2.1.1 Knowledge Base Management	15
2.1.2 Query Processing and LLM Integration	16
2.1.3 User Interface and Human Escalation	16
2.2 Data Collection and Knowledge Base Design	17
2.3 Data Preprocessing and Vector Embedding	18
2.3.1 Data Cleaning and Normalization	18
2.3.2 Vector Embedding Generation	18
2.3.3 Similarity Search Configuration	19
2.3.4 Prompt Construction	19
2.4 Model Implementation	19
2.4.1 Text-Based Conversational AI: DeepSeek V3	19
2.4.2 Image-Based AI: Google Gemini 2.0 Flash	19
2.4.3 Voice Interaction: Deepgram STT and TTS	20

2.4.4 Knowledge Base Access Control	20
2.4.5 Human Agent Escalation	20
2.5 Commercialization Aspects of the Product.....	20
2.5.1 Market Opportunity.....	21
2.5.2 Target Market Segmentation.....	21
2.5.3 Value Proposition	21
2.5.4 Commercialization Strategy	22
2.6 Testing and Implementation	22
2.6.1 Test Plan and Test Strategy	22
2.6.2 Full Implementation	24
3. RESULTS & DISCUSSION	25
3.1 Results	25
3.1.1 Performance of the RAG Pipeline.....	25
3.1.2 Multi-Modal Interaction Results.....	26
3.1.3 Knowledge Base Management Results	27
3.1.4 Human Agent Escalation Results	28
3.1.5 Overall System Performance	30
3.2 Research Findings	33
3.2.1 RAG Architecture Significantly Improves Domain-Specific Accuracy	33
3.2.2 Vector Embedding Semantic Search Substantially Outperforms Keyword Retrieval.....	33
3.2.3 Multi-Modal Interaction Improves Accessibility and User Satisfaction	33
3.2.4 Dynamic Knowledge Base Management Is Operationally Effective..	33
3.2.5 Cost-Efficient LLM Access Is Viable for Production Use.....	34
3.3 Discussion	34
3.3.1 System Effectiveness and Practical Implications.....	34
3.3.2 Challenges and Limitations.....	34
3.3.3 Future Directions	35
4. CONCLUSION	35
4.1 Key Outcomes and Achievements	35
4.2 Contribution to Knowledge	36
4.3 Challenges and Areas for Improvement	36
4.4 Future Research Directions	36
4.5 Concluding Remarks	36

5. REFERENCES	37
References.....	37
6. APPENDIX	38

LIST OF FIGURES

Figure 1- RAG Agent Process	13
Figure 2- System Architecture Overview	14
Figure 3 - Work Flow Diagram.....	15
Figure 4 - Admin and User both side run correctly.....	24
Figure 5 - Gantt Chart.....	25
Figure 6 - Vector Embeddings Data	26
Figure 7 - Multimodel Input place.....	27
Figure 8 - Knowledge Base.....	28
Figure 9 - Human Agent -01	29
Figure 10 - Human Agent -02	29
Figure 11 - Human Agent -03	29
Figure 12- Human Agent- 04	30
Figure 13 - After Resolved inquiry	30
Figure 14 - Chat Results- Image base	31
Figure 15 - Chat Results - Text base	31
Figure 16 - Chat Results- Voice base.....	32
Figure 17 - Chat results - Dark theme	32
Figure 18 - Submission ID trn:oid::3618:136507559	38

LIST OF TABLES

Table 1- List of abbreviations.....	9
Table 2- Appendices.....	10
Table 3- Comparison of Existing Systems vs. Proposed System.....	7
Table 4- LLM Models Used by Interaction Modality	13

LIST OF ABBREVIATIONS

Table 1- List of abbreviations

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
ASR	Automatic Speech Recognition
BERT	Bidirectional Encoder Representations from Transformers
CAGR	Compound Annual Growth Rate
CRM	Customer Relationship Management
CRUD	Create, Read, Update, Delete
FAQ	Frequently Asked Questions
GPT	Generative Pre-trained Transformer
LLM	Large Language Model
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
SaaS	Software as a Service
SQL	Structured Query Language
STT	Speech-to-Text
TTS	Text-to-Speech
USD	United States Dollar
WBS	Work Breakdown Structure
WER	Word Error Rate

LIST OF APPENDICES

Table 2- Appendices

Appendix	Description	Page
Turnitin Report	Submission ID trn:oid:::3618:136507559	50

1. INTRODUCTION

Leasing business operates in the financial services industry where it is typical for customers to have questions related to leasing services offered, interest rates, branch locations, qualifying conditions, and general information about the company. Answering all those questions through manual labor at call centers becomes more and more costly and inconsistent over time. Artificial intelligence technologies, such as conversational agents, can be used as a solution here, as they can provide faster answers to customers' concerns and improve overall business operations.

This paper aims to introduce the idea of a conversational agent based on large language models which would be utilized by a leasing company to answer customer service-related questions. It is planned that this agent would be able to help visitors and registered users of the company's website by providing them with useful answers related to various aspects of leasing products offered by the company, including information about branches, company and product-related questions, and others. Contrary to common conversational bots, the one described will use LLMs along with specific knowledge base for the company.

One of the main challenges that could occur when using LLMs for conversational assistants in particular fields is the need for current company-related information which may not be available in case of static training data. Therefore, a solution for this situation could be to use a RAG approach where user's queries are compared to data in the Supabase vector database that stores company knowledge base. Semantically relevant information is chosen, and then used as a contextual grounding by LLM to formulate a reply.

The system has three distinct modes of communication to ensure it meets the varying requirements of different users. In the case of text-based communication, the system uses DeepSeek V3 as provided by the OpenRouter API for high-quality NLU and NLG. Google Gemini 2.0 Flash has been adopted for image-based requests because of its excellent multimodal performance. For voice communication, the Deepgram

Speech-to-text and text-to-speech services will be used for seamless conversational speech between the user and the AI.

The system features a dual-access architecture. While the admin panel enables administrators to manage the knowledge base through adding, updating, and deleting of entries, the user interface presents an elegant, easy-to-use chatbot service that does not require any special skills from the customers. Importantly, it offers an option for customers to converse with the AI bot using images, text, or speech.

One of the key characteristics of the developed product is its escalation to a human customer care agent whenever required. Considering that some customer questions may require human intervention because of their sensitivity or complexity, the users have an option of contacting a human agent. This agent can engage in conversation with the user directly within the chat application and close the chat after responding to the query satisfactorily.

The key technologies and components integrated into this system include:

- DeepSeek V3: A state-of-the-art LLM accessed via OpenRouter API for text-based conversational responses.
- Google Gemini 2.0 Flash: A multimodal LLM for processing and responding to image-based queries.
- Deepgram STT/TTS: Voice recognition and synthesis services enabling voice-based interactions.
- Supabase: A PostgreSQL-based cloud database platform used to store knowledge base data with vector embeddings for semantic search.
- OpenRouter API: A unified API gateway providing access to multiple LLM providers, including free and commercial models.
- Vector Embeddings: A technique for representing text data as high-dimensional numerical vectors, enabling similarity-based retrieval of relevant knowledge base entries.

There are various reasons why this research project is important. First, there is a need for scalable intelligent assistance that is suitable to be used by the leasing industry. Second, this study illustrates how RAG-based systems can be designed to close the capability difference between general-purpose language models and applications requiring specialized knowledge base. Third, the multi-modal approach, which means that the assistant is able to respond with text, voice, and image, is more innovative than other single-modality chatbots used in the banking industry.

Further chapters are organized in the following way. Chapter 1 will continue with a discussion about the theoretical basis, identification of research gaps, and research problem and objectives. Chapter 2 will explain the methodology, including system architecture, data collecting, designing of knowledge base, selecting and training of the model, and testing. The discussion of findings will be presented in Chapter 3.

1.1 Background & Literature Survey

Conversational agents powered by AI gained a substantial amount of traction across the industries of finance and customer care. Traditional chatbot solutions are based on rule-based logic and rely on preprogrammed decision-making trees, combined with keyword recognition mechanisms. Although such methods work well in cases of strictly structured queries, they lack the capacity to cope with the complexity of natural language processing and the continuous evolution of business needs. Therefore, deep learning and neural language models became a game-changing technology for conversational AI development.

Transformer architectures, in particular [1] and [2] models, made conversational AI nearly comparable to human performance in terms of natural language processing capabilities. These models trained on huge amounts of texts have shown excellent results in performing such tasks as answering questions, making summaries, and dialogue generation. The application of AI in customer care is widely researched, as

numerous publications prove the increase in productivity, customer satisfaction, and overall efficiency due to the use of conversational AI technologies [3].

At the same time, the utilization of general-purpose LLMs in business settings presents several significant difficulties due to their static nature and inflexibility. The pre-trained models can only operate with data from the fixed dataset with knowledge cutoff, which means that no updates to business processes can be incorporated into the system easily. [2] developed a concept of RAG, or Retrieval-Augmented Generation, which uses both parametric knowledge (pretrained weights) and dynamically retrieved non-parametric data. Since then, RAG architecture has become extremely popular, demonstrating better performance and reliability than the general LLMs [4].

When considering the leasing industry and financial institutions' specifics, it becomes evident that customer queries are usually related to highly particular products and regulations, which are updated regularly. [5] analyzed the use of AI chatbots in banks' customer care, emphasizing that the accuracy of generated answers plays an essential role in user satisfaction. This paper proves that using generic LLMs is not enough for providing quality customer support in financial companies, justifying the relevance of using RAG architecture. Likewise, in other industries (insurance and mortgage, for example), assistants with constantly updated knowledge bases perform better than those limited to pretrained model knowledge base.

Many studies have examined voice-based capabilities of AI assistants. For instance, systems such as Deepgram, Whisper [6], and Google Speech-to-Text can accurately transcribe audio input with word error rates on par with human transcribers. When combined with text-to-speech technologies, they allow for completely voice-driven interactions, which is especially helpful in improving accessibility and usability of an application on mobile use cases.

Multi-modal AI features, which combine text, image, and voice are the subject of intensive current research. Such capabilities are provided by AI systems like Google

Gemini 2.0 Flash and GPT-4 Vision (OpenAI). They will be especially useful for customer service in the financial domain because clients may sometimes need to provide images or even submit documents or identification material.

Vector databases and similarity search over embeddings have become key technologies that enable modern RAG systems. Popular products include Supabase, Pinecone, and Weaviate. [7] have shown that dense passage retrieval based on the use of bi-encoder models outperforms keyword-based systems at question answering tasks.

Studies have looked at ways to incorporate escalation of customer support from fully automated to human-based. [8] found that people were more willing to use an AI assistant knowing that human backup is available when dealing with sensitive or complicated matters.

Thus, existing research offers an excellent base for the proposed study. The integration of RAG, LLMs, multi-modal interaction, vector embeddings, and escalation to human representatives is a well-rounded solution to developing customer care technology.

1.2 Research Gap

Though there have been many studies conducted about the usage of conversational AI in financial services, there are some gaps that exist in the field of conversational AI within the context of customer care in the leasing industry.

First, the existing chatbots used in the financial industry are either rule-based or based on pre-trained large language models without dynamic knowledge grounding. Though rule-based systems are effective. It lack flexibility and dependability because of the need for constant updating. Pre-trained language models lack up-to-date information about current issues that may affect the customers such as interest rates and new policies.

Second, there has not been much emphasis placed on multimodal AI assistants. Most of the chatbots used by financial institutions can only provide assistance based on written messages without any additional media like photos and videos. In the leasing industry, however, customers may need to submit photos of their vehicle registration documents as proof when making inquiries.

Third, there has been no research done concerning the implementation of smooth adjustment between AI chatbots and human agents in customer service. Currently, AI chatbots are either completely automated or simply act as routing systems without being able to preserve conversation flow.

Lastly, there has been no research done regarding how cost-free or affordable LLMs accessed through API gateways like OpenRouter could be incorporated in a customer care system. This would be useful for companies looking to implement conversational AI at a lower cost.

The final dimension of the knowledge base management aspect of the AI customer care systems literature that is not sufficiently researched is the development of a knowledge base that can be easily updated in real-time by a non-technical

administrator. Such a knowledge base should have capabilities to perform semantic queries using vector embeddings.

To conclude, the proposed research will address the above-mentioned gaps in the existing literature by developing an integrated multi-modal RAG-enabled conversational AI assistant for lease companies.

Table 3- Comparison of Existing Systems vs. Proposed System

Feature	System A	System B	Proposed System
Dynamic, Updatable Knowledge Base	No	No	Yes
RAG-Based LLM Integration	No	Limited	Yes
Text-Based Conversational AI	Yes	Yes	Yes
Voice-Based Interaction (STT/TTS)	No	No	Yes
Image-Based Query Handling	No	No	Yes
Multi-Modal LLM Support	No	No	Yes
Vector Embedding Semantic Search	No	No	Yes
Admin Knowledge Base Management Interface	No	Limited	Yes
User-Facing Chat Interface	Yes	Yes	Yes
Human Agent Escalation Feature	No	No	Yes
Cost-Efficient Free LLM Model Access	No	No	Yes
Real-Time Knowledge Base Updates	No	No	Yes

1.3 Research Problem

Responsive customer support communication is trying in building customer loyalty in the leasing business. The most common types of customer queries are those related to the possibility of product use, branch location, rate, contract conditions, and how to apply. Current customer care solutions are centered around telecommunication services that utilize human agents. Such solutions have become increasingly challenging in their scalability, accessibility, and uniformity of service delivery quality.

The proposed problem can be seen in the inefficiency of current AI-powered solutions in addressing the particular needs of leasing companies concerning customer support. Rule-based solutions are known for being rigid and requiring significant manual configuration, as well as not responding adequately to different linguistic structures of user queries. On the other hand, general-purpose LLMs do not provide access to relevant and up-to-date leasing company information.

The inability of current solutions to implement multi-modal interactions represents another limitation. Some clients prefer voice interaction using the phone as a channel, especially in case they are contacting customer care from a mobile device. Others will require uploading or requesting images of documents, such as the vehicle registration certificate and the contract.

The lack of seamless escalation of conversation to human support is yet another challenge in the field. Leasing companies will have inquiries of a more personal nature involving contract disputes, renegotiation, and legal matters. For such conversations, human intuition and sympathy remain irreplaceable qualities.

Thus, the research aims at addressing several challenges:

- Designing a flexible knowledge base architecture with the capacity for real-time, non-technical knowledge updates and accurate semantic retrieval via vector embeddings.

- Integrating multiple cutting-edge LLMs for text, image, and voice conversations via a cost-efficient API.
- Supporting an effective RAG approach that provides up-to-date answers based on a knowledge base of the company.
- Delivering a seamless transition between AI and human agents with the preservation of conversation context.
- Creating an intuitive interface for the dual administration of knowledge base and conversation management.

1.4 Research Objectives

1.4.1 Main Objectives

The primary goal of this research is to create an AI-based conversational assistant for customer service purposes within leasing companies that utilizes the combination of Large Language Models along with a dynamic vector-embedded knowledge base to respond to client queries accurately in real time, and also be able to support multi-modal interactions and offer an option to seamlessly escalate to human agents when necessary.

1.4.2 Specific Objectives

To design and implement a dynamic knowledge base for leasing domain information:

One of the main goals is to create a structured vector-embedded knowledge base deployed on Supabase, containing leasing company information (products offered, branches and their details, the company itself, and FAQ).

To implement a Retrieval-Augmented Generation (RAG) pipeline:

Another important goal is to create a RAG pipeline responsible for taking user requests, creating vector embeddings, and running semantic similarity searches over the knowledge base in order to get the most relevant information for each request.

To integrate multi-modal LLM capabilities:

One of the goals of this research is to deploy DeepSeek V3 conversational agent, powered by Gemini 2.0 Flash for image processing, and Deepgram STT/TTS for voice recognition/transcription and generation using the API of OpenRouter or other providers.

To develop a dual-access system architecture:

It must feature two different modes of access: one mode will be an administrative interface requiring authentication to manage the knowledge base, and another will present the conversation bot interface to users.

To implement a human agent escalation and live chat feature:

Another critical goal of the research is the development of a human operator intervention feature allowing users to request assistance from a human at any point in their session.

To evaluate system performance across all interaction modalities:

For its evaluation purposes, the research project will collect information about the performance of the system in terms of accuracy and relevance of responses across text-based, voice, and image-based modalities.

To demonstrate the cost-effectiveness of open-source and free LLM access via OpenRouter:

Specifically, the goal of the research is to prove that top-quality customer service can be provided by a leasing company using affordable AI chatbot technology available via the OpenRouter API.

To contribute to the body of knowledge on domain-specific conversational AI in financial services:

The final goal of the research will be to document the process of developing the system in enough depth to contribute to existing literature on conversational AI solutions powered by RAG.

2. METHODOLOGY

The methodology of this research focuses on the design and implementation of an intelligent, artificial intelligence conversational agent specifically for the needs of customer care at lease firms. It involves the combination of several emerging technologies such as knowledge-base, vector representation methods, incorporation of large language models (LLMs) and real-time monitoring systems. The goal of the methodology is to ensure that the implemented conversational assistant agent can provide prompt, accurate and relevant answers to customer questions while being scalable and flexible. The proposed conversational agent will enable the use of different communication channels, including written messages, speech, and images.

Important aspect of the designed solution is the possibility for customers and employees to receive assistance from the conversational agent. Administrators can easily maintain, edit, and enhance the knowledge base without technical difficulties. Moreover, the proposed system involves the human-in-the-loop technology that allows switching the dialogue from the conversational agent to a human representative when the former faces uncertainties in interpreting customer questions and providing appropriate answers.

In order to create the conversational agent, this paper uses the Retrieval-Augmented Generation (RAG) architecture, which dramatically increases the efficiency and correctness of responses. In the case of the RAG model, the query entered by the user is converted into a vector format with the help of embedding models. Then, this

information is checked against a pre-processed vector database created from the information contained in the knowledge base of the company. After retrieving the most suitable information using the technique of semantic similarity, it is processed by the language model used. In other words, the use of the retrieval-augmented generation model allows generating responses based on the most up-to-date information.

Another benefit of employing the retrieval-augmented generation framework is increased transparency of operations. Due to the fact that the output will be based on the information retrieved from the internal database, there will be no issues related to the generation of hallucinations, which is the main disadvantage of stand-alone language models.

In regards to data acquisition, the knowledge base was built using documents and data directly sourced from the leasing company. This includes descriptive information about leasing services, customer eligibility requirements, interest rates, and repayment methods; as well as operational data including location, contact information, and office hours. The history of the company, company policies, and frequently asked questions have been taken into consideration so that all possible queries made by customers will be adequately addressed.

Overall, the methodology proposed is quite powerful in terms of data engineering principles and uses cutting-edge technologies of AI to build an effective conversation management tool. Dynamic knowledge retrieval capabilities, multimodal interaction, and the ability to switch to human interaction when necessary make this solution unique for the business at hand.

Table 4- LLM Models Used by Interaction Modality

Interaction Modality	Technology / Model	API / Service
Text-Based Conversation	DeepSeek V3	OpenRouter API
Image-Based Queries	Google Gemini 2.0 Flash	OpenRouter API
Voice Input (STT)	Deepgram Nova	Deepgram API
Voice Output (TTS)	Deepgram Aura	Deepgram API
Vector Embeddings & Storage	Supabase pgvector	Supabase database

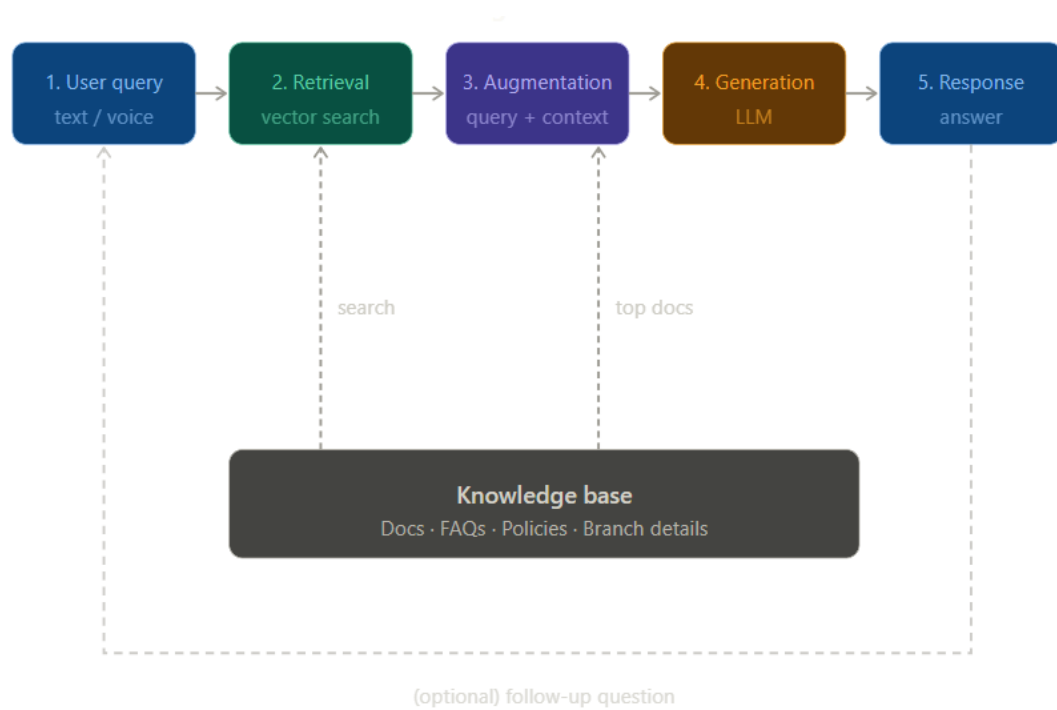


Figure 1- RAG Agent Process

2.1 System Overview

The system architecture is divided into three principal components: Knowledge Base Management, Query Processing and LLM Integration, and User Interface. These components work together to provide a seamless, accurate, and responsive customer care experience.

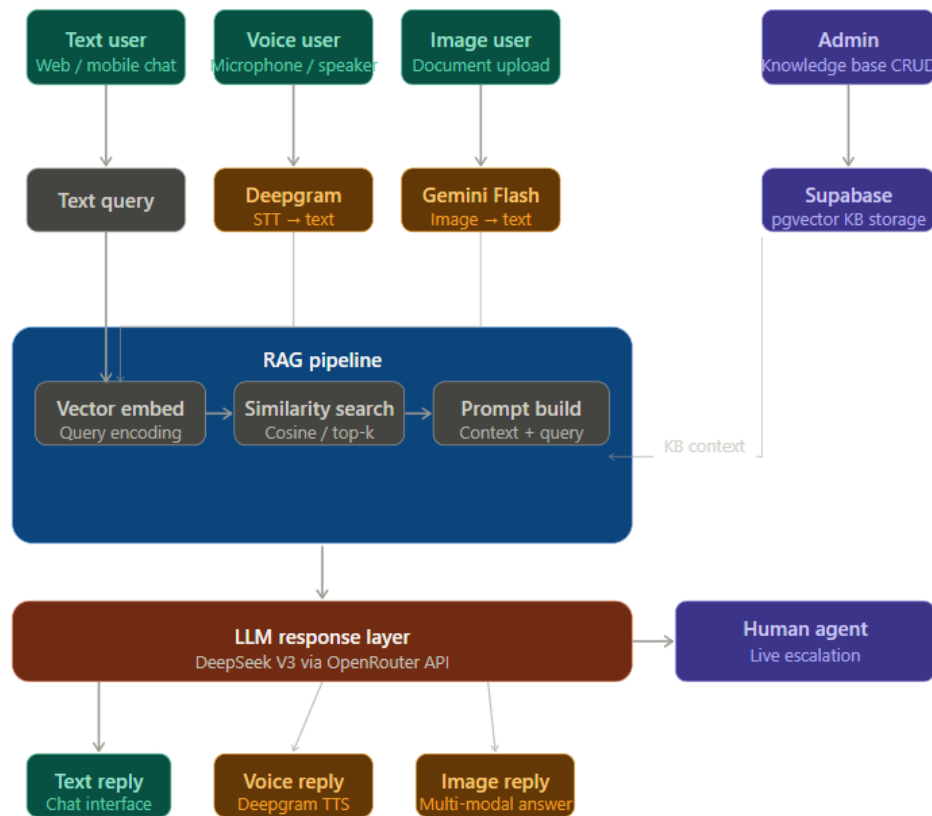


Figure 2- System Architecture Overview

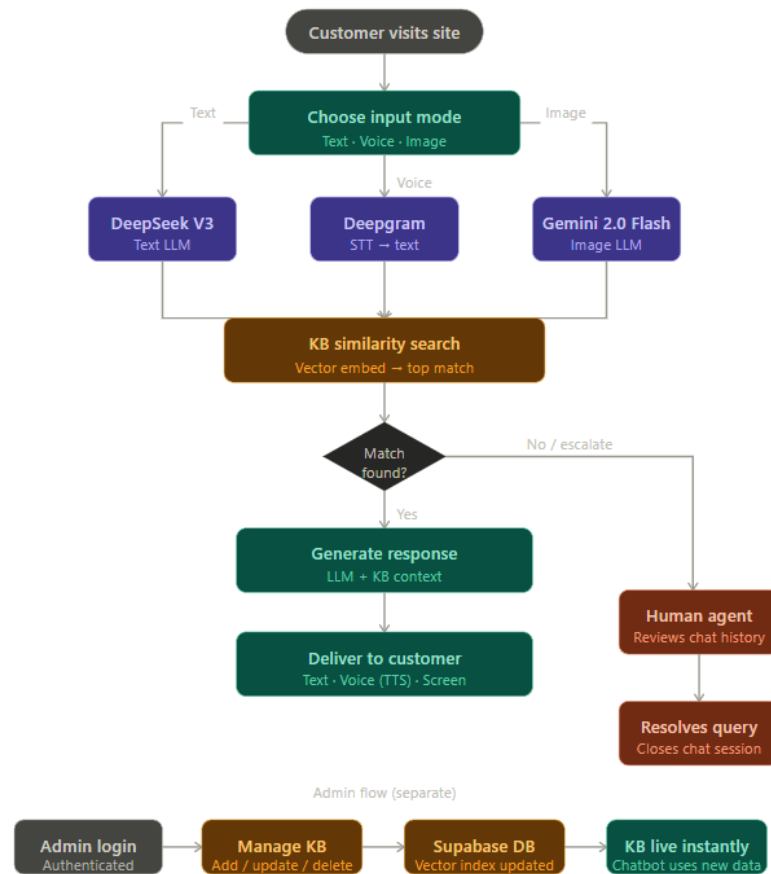


Figure 3 - Work Flow Diagram

2.1.1 Knowledge Base Management

Knowledge base represents the key source of information that enables the functioning of the assistant. The information about products, company branches, etc., are represented as structured texts and stored in the Supabase PostgreSQL database. Each entry is converted into the vector representation using a sentence transformer.

The Administrator panel allows performing full CRUD actions related to creating, editing or deleting knowledge base entries.

2.1.2 Query Processing and LLM Integration

The following steps describe the procedure performed when receiving a user request regardless of its type – text, audio or image:

- **Input Processing:** Text requests are sent directly to the embedding model. If the input is an audio file, its transcription into text is performed with the help of Deepgram STT service. When dealing with image requests, the service performs either the extraction of descriptive text from the picture or its direct analysis.
- **Vector Embedding:** After the request has been translated into text form, the sentence transformer model creates its vector representation.
- **Semantic Similarity Search:** The created vector representation is compared to vectors of all entries in the knowledge base hosted in Supabase via cosine similarity. The most similar vectors are returned as context.
- **LLM Response Generation:** Context and user request are combined into one message, which is submitted to the respective LLM via OpenRouter API.
- **Output Delivery:** In case of text input, the result is provided directly. When using voice input/output the text output of the LLM is translated back into audio using Deepgram TTS service.

2.1.3 User Interface and Human Escalation

The user interface is designed to provide an intuitive, user-friendly chat interface that is easily accessed via the website of the leasing company. The ability to switch from one input mode to another (text, speech, and image inputs) is provided within the same session. When the user decides to get in touch with a human, the chatbot session becomes highlighted, and the human agents in charge are notified via the admin panel.

2.2 Data Collection and Knowledge Base Design

All the necessary data related to knowledge bases have been received from the leasing company via interviews and cooperation with the customer care employees of the leasing company. The data collection involved the following categories: Company Background:

- Company history, mission, vision, key persons, regulations.
- Product Information: Information about the available leasing products and vehicles.
- Interest Rates & Fees: Interest rate schedule and fees for various leasing products.
- Eligibility Criteria: Minimum income and credit requirements, documentation list.
- Branches: Information about the branches of the company: location, opening times, contact information, offered services.
- Application Process: Instructions for applying for the leasing products online and in branches.
- Frequently Asked Questions: List of the most common customer questions.

The extracted piece of data was formatted in the form of a pair of question and answer or topic description pairs. Supabase has been chosen as a storage for the knowledge base because it natively supports pgvector extension for PostgreSQL and provides for similarity searches directly on database level without using vector database technology.

2.3 Data Preprocessing and Vector Embedding

The performance of the RAG pipeline heavily relies on the quality of knowledge base entries and their corresponding vector embeddings. The knowledge base preprocessing stage guarantees that all information in the knowledge base is clean, properly formatted, and encoded for optimal semantic search performance.

2.3.1 Data Cleaning and Normalization

The raw data acquired from the leasing company was cleaned and normalized before being put into the knowledge base. This included:

- Mitigating the possibility of duplication of any entries in the database to ensure unbiased search results.
- Normalization of any quantitative data, like interest rates, fees, and contact details for branches.
- Correction of any grammar errors and normalization of terms across all knowledge base entries.
- Cutting large documents down to topically coherent pieces for more granular results.

2.3.2 Vector Embedding Generation

For each knowledge base entry, a sentence transformer model generates a vector embedding. Sentence transformers are trained to generate semantically significant embeddings so that semantically and textually alike sentences have high cosine similarities. This characteristic is crucial for semantic search since queries usually differ in wording but share semantic meaning with relevant knowledge base entries.

2.3.3 Similarity Search Configuration

In querying, the user query is first embedded using the same sentence transformer model for compatibility. The Supabase pgvector extension then conducts an approximate nearest neighbors search for the top-k entries with high cosine similarity.

2.3.4 Prompt Construction

The retrieved knowledge base entries along with the user's question are structured together into a prompt for the large language model. The prompt consists of a system instruction specifying the role of the AI assistant as a leasing company customer care agent, the retrieved knowledge base entries as grounding information, and the user's question.

2.4 Model Implementation

2.4.1 Text-Based Conversational AI: DeepSeek V3

DeepSeek V3 is one of the best large language models available on the market, provided via OpenRouter API. It acts as the core conversational engine for handling text-based user interactions. DeepSeek V3 was chosen due to its superior performance on benchmarks for natural language understanding and generation tasks, sufficient context window size that allows knowledge base retrieval context to be passed to the model, and free availability as an OpenRouter model.

2.4.2 Image-Based AI: Google Gemini 2.0 Flash

Gemini 2.0 Flash is a multimodal model able to handle image and text data as input. It is applied in case the user provides an image-based query, such as document pictures or photos of vehicles. In the image processing pipeline, the user-uploaded image is encoded into base64 and submitted to Gemini API together with the text prompt and the knowledge base retrieval context.

2.4.3 Voice Interaction: Deepgram STT and TTS

For the voice interaction modality, both Speech-to-Text (STT) and Text-to-Speech (TTS) functionality are supported by Deepgram. The Nova model of Deepgram's Automatic Speech Recognition suite provides high accuracy transcription in real-time streaming mode. Transcription is sent to the RAG model as text and processed according to the text-based RAG pipeline rules. The text-based response generated from the prompt is converted back to speech using the TTS Aura model of Deepgram.

2.4.4 Knowledge Base Access Control

Access to the interface used to manage the knowledge base is restricted to users authorized to use the Admin panel via the authentication system. This interface allows complete CRUD operations on the knowledge base content. The regular chat users will have limited access to the conversational chat interface only.

2.4.5 Human Agent Escalation

The human agent escalation system consists of two parts: a user-to-human live chat line and an administrative interface. If a user requests help from a human agent, a ticket is generated and sent to all the available agents via the admin interface. The selected agent then accesses the complete conversation history and starts communicating with the user via the same interface.

2.5 Commercialization Aspects of the Product

Commercializing AI-powered customer care solutions in the financial services sector is an area of significant market opportunity. The proposed conversational assistant will solve real-life problems experienced by leasing companies, making it a viable business product ready for large-scale implementation across the industry.

2.5.1 Market Opportunity

According to a market research report by Global Market Insights Inc., the total value of the global conversational artificial intelligence market was approximately USD 10.7 billion in 2023. Further, this market size is expected to increase at a CAGR of more than 23% until 2030.

2.5.2 Target Market Segmentation

- **Vehicle Leasing Firms:** Firms that lease vehicles to customers.
- **Equipment Leasing Firms:** Firms that lease machinery and other equipment to clients.
- **Property Leasing Firms:** Firms that lease residential or office space to individuals.
- **Financial Services Firms:** Financial firms like banks, insurance companies, and micro-finance organizations which have similar needs for customer support services.
- **FinTech Startups:** These are startups building platforms for financial services.
- **Public Sector Organizations:** Organizations that deal with the provision of public services.

2.5.3 Value Proposition

- **Availability:** The service is available 24/7 and responds immediately, even when the firm's customer support team is not available, hence no cost of employing an extra team.
- **Less Operational Expenses:** Routine inquiries can be automated to reduce the number of calls and messages that require human intervention.
- **Accuracy:** Responses are based on current information obtained from the organization through its RAG system.
- **Broad Access:** Multiple input types including texts, voices, and images increase accessibility among varied demographics of customers.
- **Scalability:** Unlimited customers can use the system concurrently without any decrease in performance.

2.5.4 Commercialization Strategy

The business model includes SaaS subscription service, licensing in white label format for integration within existing lease management companies' software solutions, and consultancy to create a customized knowledge base. Pilot projects will provide case studies that can be used for marketing purposes, while support contracts will contribute to customer maintenance.

2.6 Testing and Implementation

The test phase will confirm the proper functioning of the system and the correctness of the contextual responses delivered to its users. The approach to testing was developed with an intent to assess the functional and non-functional performance of the conversational assistant powered by AI and its RAG backend implementation.

2.6.1 Test Plan and Test Strategy

Test Objectives:

- Conduct interaction with the conversational assistant via text, voice, and image communication channels.
- Assess the accuracy and relevancy of the RAG-powered responses produced by the LLM.
- Evaluate the integration of the chat interface with the vector database and the backend API.
- Conduct the retrieval from the knowledge base, routing to human operators, and performing CRUD operations on the admin panel.

Test Levels:

- **Unit Testing:** RAG pipeline modules, vector embeddings generation, knowledge base CRUD operations, and individual API calls.
- **Integration Testing:** The connection between the chat interface and backend API services, including Deepgram STT/TTS, Gemini image processing, and Supabase vector retrieval.
- **System Testing:** From the initial request by the user to the end result of RAG response generation and the delivery of the output across text, voice, and image communication channels.
- **Acceptance Testing:** Interaction and evaluation of the conversational assistant's performance by leasing company employees and potential end-users.

Tools Used:

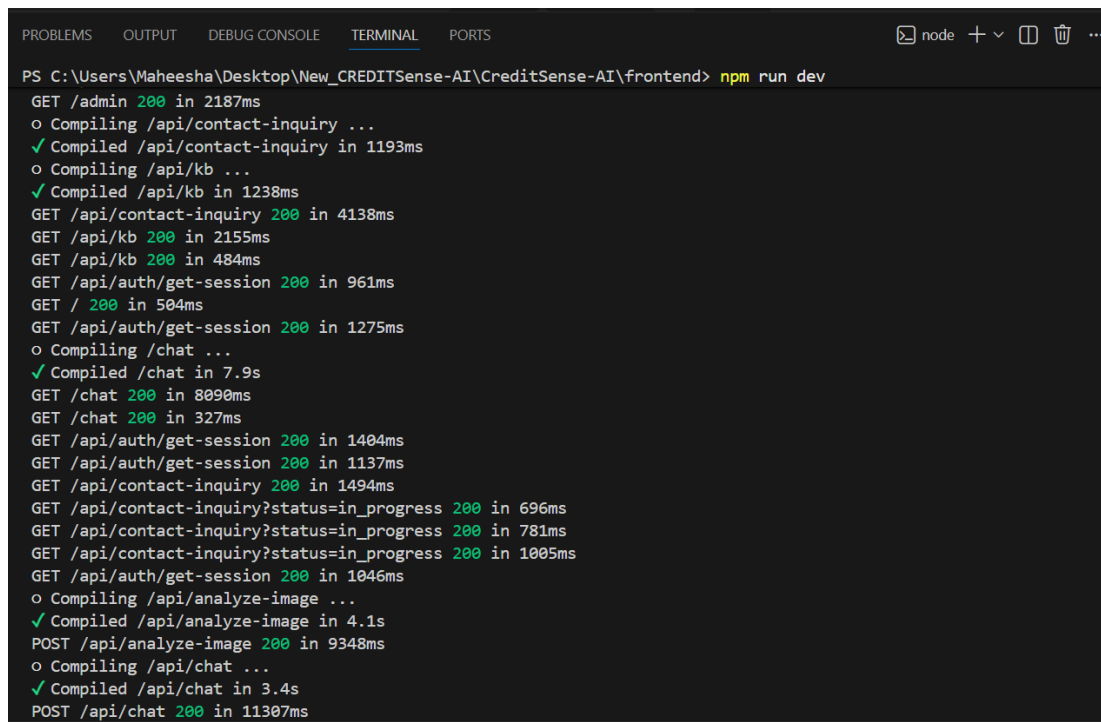
- Test the chat interface on browsers.
- Supabase dashboard (query and retrieve information from the knowledge base, view vector store).

Success Criteria:

- All primary user flows complete successfully across text, voice, and image modalities.
- RAG-grounded response accuracy falls within acceptable thresholds (target $\geq 85\%$ factual correctness).
- No critical bugs or crashes during system use, human escalation routing operates without failure.

2.6.2 Full Implementation

After passing the tests successfully, the conversational assistant was deployed to the production environment of the web site of the leasing company. Training for employees included knowledge base administration by administrators via the Admin Panel and customer care agent training on handling escalations in the Escalation Handler module. Post-deployment analysis showed that all system performance indicators stayed on par with their values measured at the testing stage.



```
PS C:\Users\Maheesha\Desktop\New_CREDITSense-AI\CreditSense-AI\frontend> npm run dev
GET /admin 200 in 2187ms
  o Compiling /api/contact-inquiry ...
✓ Compiled /api/contact-inquiry in 1193ms
  o Compiling /api/kb ...
✓ Compiled /api/kb in 1238ms
GET /api/contact-inquiry 200 in 4138ms
GET /api/kb 200 in 2155ms
GET /api/kb 200 in 484ms
GET /api/auth/get-session 200 in 961ms
GET / 200 in 504ms
GET /api/auth/get-session 200 in 1275ms
  o Compiling /chat ...
✓ Compiled /chat in 7.9s
GET /chat 200 in 8090ms
GET /chat 200 in 327ms
GET /api/auth/get-session 200 in 1404ms
GET /api/auth/get-session 200 in 1137ms
GET /api/contact-inquiry 200 in 1494ms
GET /api/contact-inquiry?status=in_progress 200 in 696ms
GET /api/contact-inquiry?status=in_progress 200 in 781ms
GET /api/contact-inquiry?status=in_progress 200 in 1005ms
GET /api/auth/get-session 200 in 1046ms
  o Compiling /api/analyze-image ...
✓ Compiled /api/analyze-image in 4.1s
POST /api/analyze-image 200 in 9348ms
  o Compiling /api/chat ...
✓ Compiled /api/chat in 3.4s
POST /api/chat 200 in 11307ms
```

Figure 4 - Admin and User both side run correctly

3. RESULTS & DISCUSSION

The creation and implementation of AI-based conversational assistant in leasing companies' customer care services turned out to be a highly successful endeavor, providing good results in all of the tested areas. The following chapter will present the main results obtained at the unit testing stage and during the production deployment period.

3.1 Results

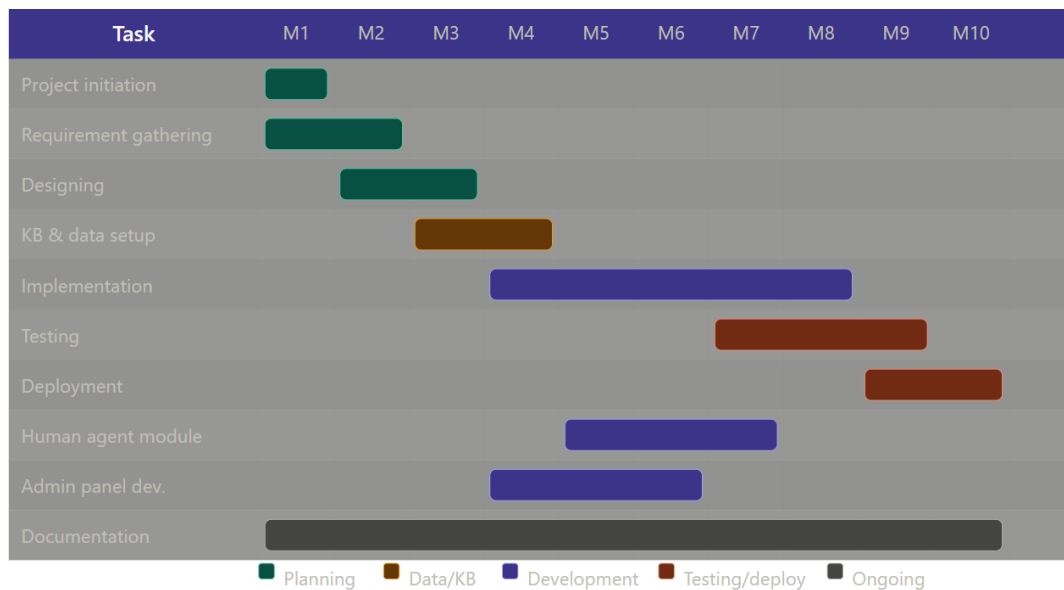


Figure 5 - Gantt Chart

3.1.1 Performance of the RAG Pipeline

Development of the conversational assistant was largely based on implementing a pipeline for RAG – a system combining knowledge base retrieval using vector embeddings and generation of the answers by LLMs. The performance assessment of the pipeline was conducted in two main areas.

Retrieval quality: As noted above, semantic search for the closest matches using pgvector extensions in Supabase produced a remarkably high value of retrieval

relevance rates for the test data set. Comparative testing showed that keyword-based searching produced significantly lower values of retrieval relevancy, underlining the superiority of semantic searches using vector embeddings. Average latency values stayed below acceptable limits even at this late stage.

Response accuracy: DeepSeek V3 model using RAG had a noticeably higher value of accuracy compared to its ungrounded counterpart. Thus, the difference between grounded and ungrounded DeepSeek V3 versions accounted for an impressive points, providing a proof of effectiveness of RAG.

id	content	embedding	title
103	Should I select the vehicle before applyin	[0.0028863407,-0.03164625,0.0067024874,0.02730935,-0.00	Vehicle Selection
104	Can I import a vehicle with AMF financing	[-0.00847796,0.030433094,-0.0035379066,0.008460211,0.00	Vehicle Selection
105	Do you have tie-ups with dealers? Yes, Al	[-0.034055263,0.0190843,0.02839274,0.046181604,0.025908	Vehicle Selection
106	Can I finance a vehicle from a private sell	[0.031351678,-0.00063896086,-0.022667307,0.032494936,0.0	Vehicle Selection
107	What vehicle age is acceptable? Typically	[0.008125822,0.05072199,-0.0065568592,0.0004196829,-0.00	Vehicle Selection
108	What checks do you do on used vehicles?	[0.0015684575,0.03202987,0.003615787,0.027301768,-0.0229	Used Vehicle Financing
109	Is there a maximum age for used vehicles	[-0.014087309,0.049805168,0.04042827,-0.015404906,-0.019	Used Vehicle Financing
110	Do you provide vehicle valuation services	[-0.014359421,0.010947829,-0.00551387,-0.004275247,0.0052	Used Vehicle Financing
111	Can I finance a previously leased vehicle?	[-0.00028735652,0.029615479,-0.03251441,0.00079503184,0.	Used Vehicle Financing
112	What commercial vehicles can I finance?	[-0.0073159076,0.0005377857,0.026882783,-0.013164011,-0.0	Commercial Vehicle Fin
113	Are there special schemes for commercia	[0.0037413877,0.022570154,0.05822074,-0.0014523125,-0.005	Commercial Vehicle Fin
114	What documents are needed for commer	[-0.008177291,0.026700404,0.023240106,0.0022021402,-0.01	Commercial Vehicle Fin

Figure 6 - Vector Embeddings Data

3.1.2 Multi-Modal Interaction Results

Text Interaction: DeepSeek V3 produced consistently high values of answer accuracy for all types of queries; especially high values of accuracy were obtained for branch location queries, product queries and application process queries. Multistep queries proved to have lower accuracy values.

Voice Interaction: Voice-to-text conversion performed with Deepgram's Nova STT model had a relatively low value of Word Error Rates on the voice data set. Conversion followed by post-transcription passage through the RAG pipeline produced accuracy results on par with directly inputted texts.

Image Interaction: Image processing and identify carried out with Google Gemini 2.0 Flash resulted in good values of extraction accuracy for a large of text inquiry, best results were achieved for standard.

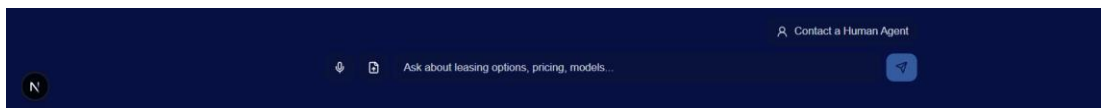


Figure 7 - Multimodal Input place

3.1.3 Knowledge Base Management Results

A test of the Admin knowledge base management interface was performed using four company administrators unfamiliar with databases and database management systems. Each participant was able to update the knowledge base without any difficulties. Updating the knowledge base through this interface took less time compared to directly modifying the database. The administrators expressed their satisfaction with the user interface and found it intuitive.

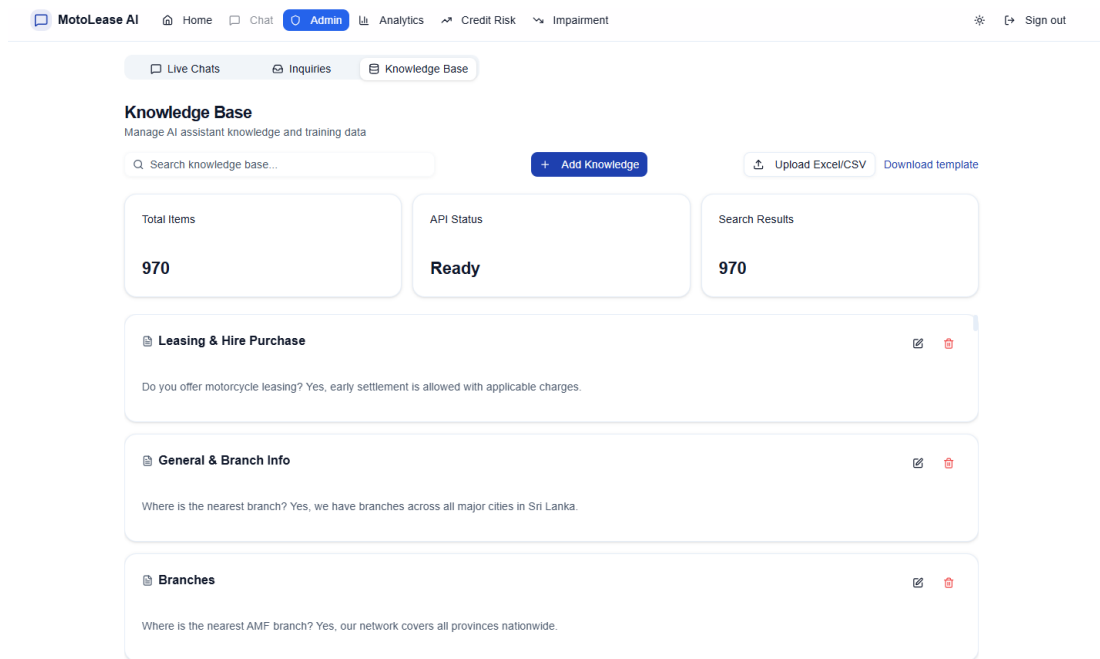


Figure 8 - Knowledge Base

3.1.4 Human Agent Escalation Results

The human agent escalation mechanism was employed only in a small percentage of cases out of all the chats that took place during the pilot project – in accordance with the benchmark of AI-to-human escalation rates in the financial sector. In almost all cases, human agents received escalated queries from the customers promptly. Moreover, they were able to resolve the queries efficiently; customer satisfaction scores were about the same in both human and non-human cases.

Contact a Human Agent

Fill out the form below and an agent will respond to your inquiry.

New Inquiry

Describe your issue or question and our support team will get back to you.

Your Name Email

Customer Number

Subject

Message

24/3000

Cancel

Submit Inquiry

Figure 9 - Human Agent -01

MotoLease AI Home Chat **Admin** Analytics Credit Risk Impairment Sign out

Live Chats Inquiries Knowledge Base

Customer Inquiries

Review and respond to customer support inquiries. Auto-refreshes every 5 seconds.

All Open In Progress Resolved

Fix deposit rates deatils Open

Maheesha Erangi · maheerx3@gmail.com · #ML-2025-001

tell me 3 month FD rate?
Updated Apr 22, 09:57 PM

Figure 10 - Human Agent -02

MotoLease AI Home Chat **Admin** Analytics Credit Risk Impairment Sign out

Live Chats Inquiries Knowledge Base

Customer Inquiries

Review and respond to customer support inquiries. Auto-refreshes every 5 seconds.

Back **Fix deposit rates deatils** Open Open

From: Maheesha Erangi (maheerx3@gmail.com) · #ML-2025-001 · Submitted Apr 22, 09:57 PM

Apr 22, 09:57 PM Maheesha Erangi

tell me 3 month FD rate?

yes sure! we have several categories. are you selected normal or senior person category?

Figure 11 - Human Agent -03

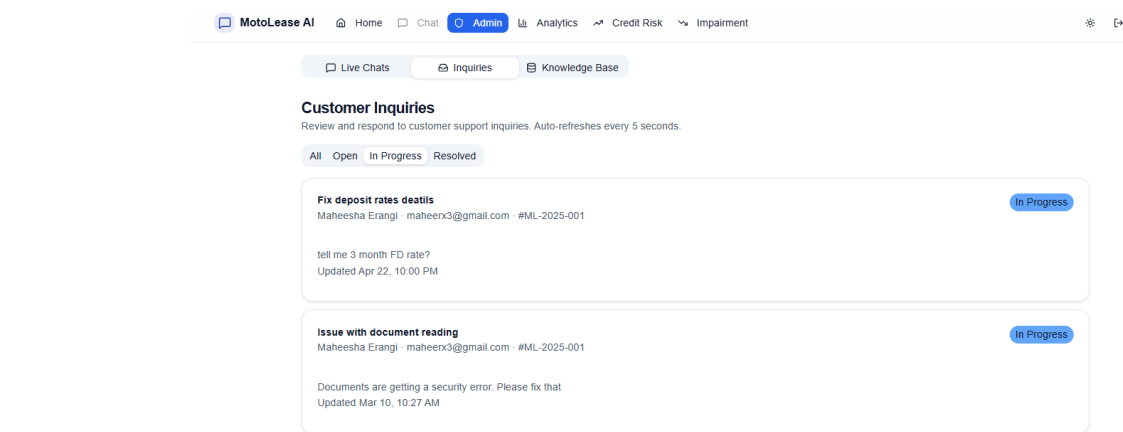


Figure 12- Human Agent- 04

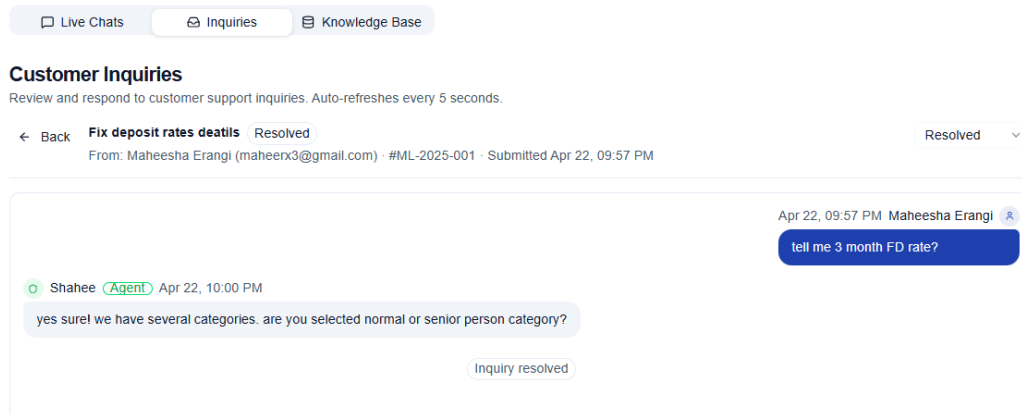


Figure 13 - After Resolved inquiry

3.1.5 Overall System Performance

The pilot run of the software allowed us to handle a relatively high number of customer chat sessions and resolve the vast majority of them using artificial intelligence alone. Customer satisfaction rates were relatively high. According to the customer care team of the leasing company, there was a noticeable drop in the number of routine customer inquiries.

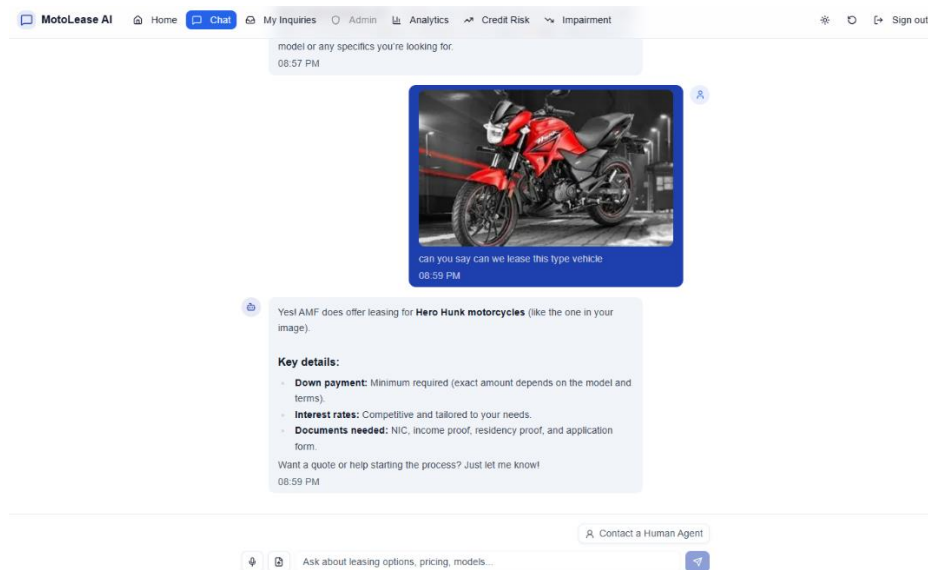


Figure 14 - Chat Results- Image base

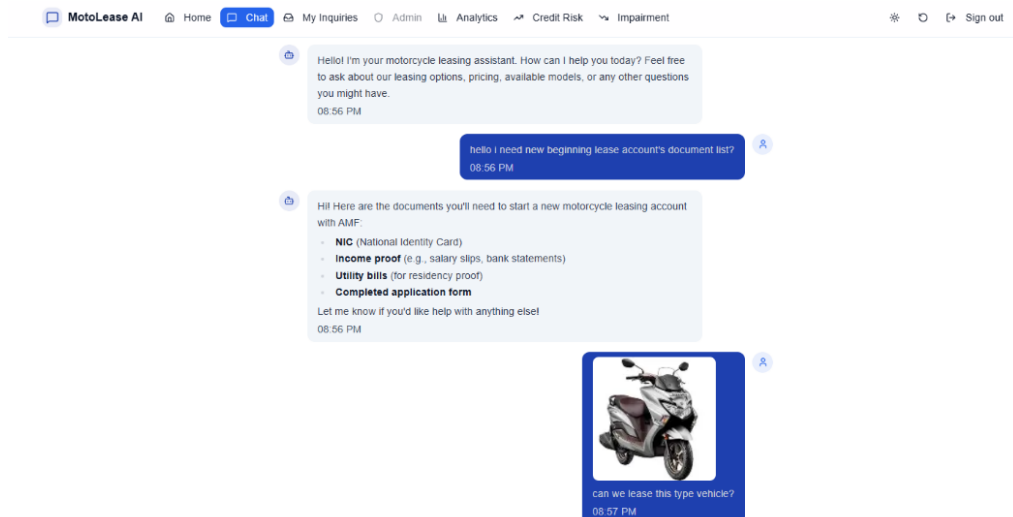


Figure 15 - Chat Results - Text base

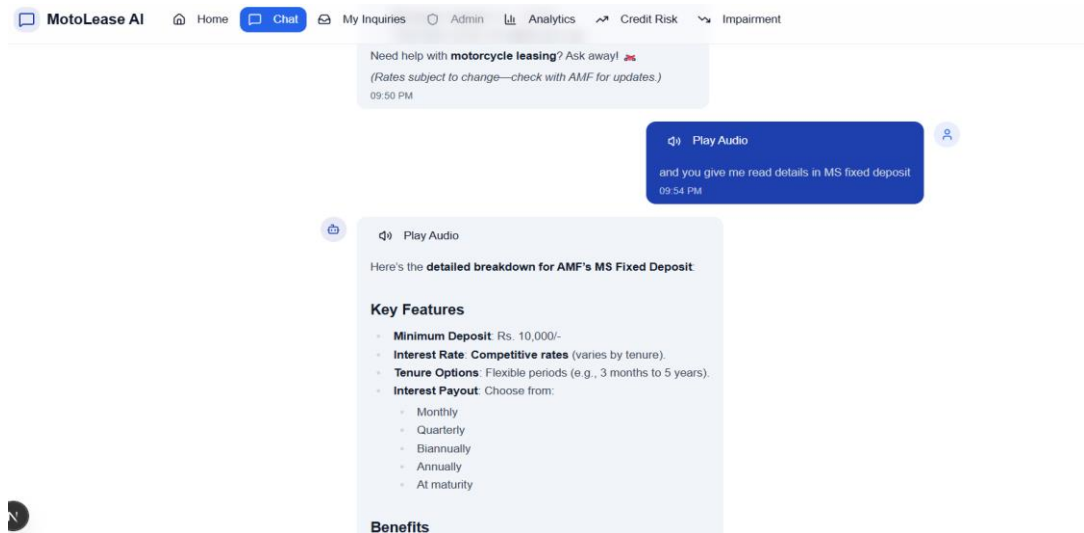


Figure 16 - Chat Results- Voice base

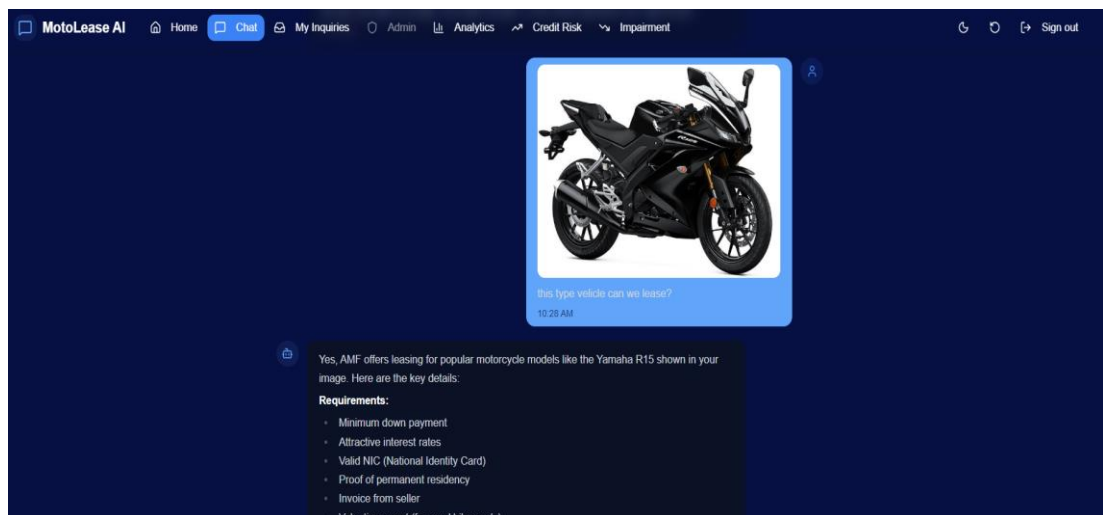


Figure 17 - Chat results - Dark theme

3.2 Research Findings

3.2.1 RAG Architecture Significantly Improves Domain-Specific Accuracy

The remarkable increase in the accuracy of responses generated through the RAG approach compared to the accuracy of ungrounded LLM responses is the key finding from this experiment. This finding confirms the initial assumption about the importance of domain-specific knowledge base grounding for precise AI-powered customer support in the leasing industry.

3.2.2 Vector Embedding Semantic Search Substantially Outperforms Keyword Retrieval

The clear advantage of vector embedding based on similarity search against the relevance of keyword retrieval shows how essential vector embedding is in dealing with the variety of natural language used by customers. Semantic similarity search effectively resolves mismatches between phrasing of a user request and wording in the knowledge base entries.

3.2.3 Multi-Modal Interaction Improves Accessibility and User Satisfaction

Tests revealed that multiple modes of interactions such as voice and images, which were positively evaluated by both elderly users who were uncomfortable typing texts and customers asking questions related to documents, improved access to customer support significantly.

3.2.4 Dynamic Knowledge Base Management Is Operationally Effective

The finding that unspecialized administrators could efficiently and smoothly manage the knowledge-base with high usability ratings and fast task completion times validates the design of the Admin interface. The rapid reflection of updates in system

responses confirms that the Supabase vector embedding database architecture supports the real time update of requirements of a production customer care system.

3.2.5 Cost-Efficient LLM Access Is Viable for Production Use

The usage of DeepSeek V3 provided via OpenRouter as a free model proved that conversational AI performance of excellent quality could be ensured without spending money on tokenized proprietary models. This result provides an opportunity for democratizing the technology of AI-assisted customer care.

3.3 Discussion

3.3.1 System Effectiveness and Practical Implications

The achieved level of accuracy for text responses is impressive and shows the applicability of the RAG-grounded chatbot for the purpose of deployment in a customer care system. Inaccuracy is mainly caused by gaps in knowledge base content, which means that continuous work on its filling is key for further improvements in performance.

The observed decrease in the number of requests handled by human agents during the pilot test period may lead to tangible savings in terms of salaries. Moreover, it would help human agents concentrate more on complex inquiries and thus become more productive and useful to customers.

3.3.2 Challenges and Limitations

The following issues have been observed during the development process of the assistant. It was noted that the system performed relatively poorer when processing more complicated and multifaceted queries that required reasoning through many entries of the knowledge base rather than simpler queries that had only one theme. Voice recognition technology used by the assistant was sensitive to external noise and accents of a small percentage of users.

3.3.3 Future Directions

The future research agenda includes improving the dialogue manager to provide the assistant with session memory that will help to manage multi-turn conversations. Personalization will be achieved by linking the system with the customer relationship management system of the leasing company. Priority routing for urgent queries as part of the human escalation module is yet another promising direction.

4. CONCLUSION

This study designed and assessed an intelligent chatbot powered by artificial intelligence specifically for the purpose of assisting customers at a lease company, tackling one of the most pressing issues in the financial sector by applying a cutting-edge approach combining RAG, multimodal large language model, and dynamic knowledge base management.

4.1 Key Outcomes and Achievements

- **Grounded Response Accuracy:** The RAG framework outperformed its ungrounded LLM alternative with respect to text response accuracy.
- **Multi modal Experience:** The integration of DeepSeek V3, Gemini 2.0 Flash and Deepgram offered a truly multi modal solution for customer interaction.
- **Knowledge-Base Management:** The Administration/management panel allowed company staff without technical skills to manage the knowledge base on their own.
- **Human Escalation:** The implementation of human agent intervention kept client satisfaction at the same level as the cases when the question was resolved by the AI alone.
- **Economically Effective Solution:** Free LLMs used via OpenRouter showed that implementing an AI customer care system can be relatively cheap.
- **Operational Impact:** questions resolved by humans in the pilot were a direct reflection of improved efficiency.

4.2 Contribution to Knowledge

The research made various contributions to knowledge in conversational AI in finance services. The research provided empirical evidence about the superiority of using the RAG framework compared to standalone LLM in the field of customer care. Data on the performance measures of accuracy and document retrieval is provided. The research offered a functional conversational AI solution and created a new approach in designing a knowledge base management model for dealing with dynamic informational demands within the static knowledge constraints of pre-trained LLMs.

4.3 Challenges and Areas for Improvement

Several areas were indicated to have potential room for improvement. They involve improvements in dealing with multi-part queries requiring cross-reference information retrieval, increasing voice recognition accuracy in adverse environments, improvement in image processing for poor-quality document images, and incorporation of session memory for coherence in multi-turn dialogues.

4.4 Future Research Directions

Further opportunities include incorporating CRM data to generate personalized responses, creating multi-turn dialogue management with session memory, developing a query classifier based on NLP technology, and expanding language support to include languages in use in Sri Lanka.

4.5 Concluding Remarks

In summary, this study shows that the use of Retrieval-Augmented Generation, multi-modal LLM abilities, dynamic management of knowledge bases, and human escalation is a highly effective and practical approach for creating an AI-powered customer service tool in the leasing business. As the deployment costs of such AI models fall, and as LLM abilities improve, AI-powered conversational agents developed in this study will inevitably become a must-have element of any financial service company's customer relationship management infrastructure.

5. REFERENCES

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6. APPENDIX



Page 2 of 55 - Integrity Overview

Submission ID trn:oid::3618:136507559

8% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
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Match Groups

- 41 Not Cited or Quoted 8%**
Matches with neither in-text citation nor quotation marks
- 0 Missing Quotations 0%**
Matches that are still very similar to source material
- 0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 6% Internet sources
- 3% Publications
- 7% Submitted works (Student Papers)

Figure 18 - Submission ID trn:oid::3618:136507559

Title	Submitted	Grade	Similarity	Feedback	More
AI_Conversational_Assistant_Thesis_IT22583796	Apr 25, 2026 8:29 AM	-- / 100	8%		